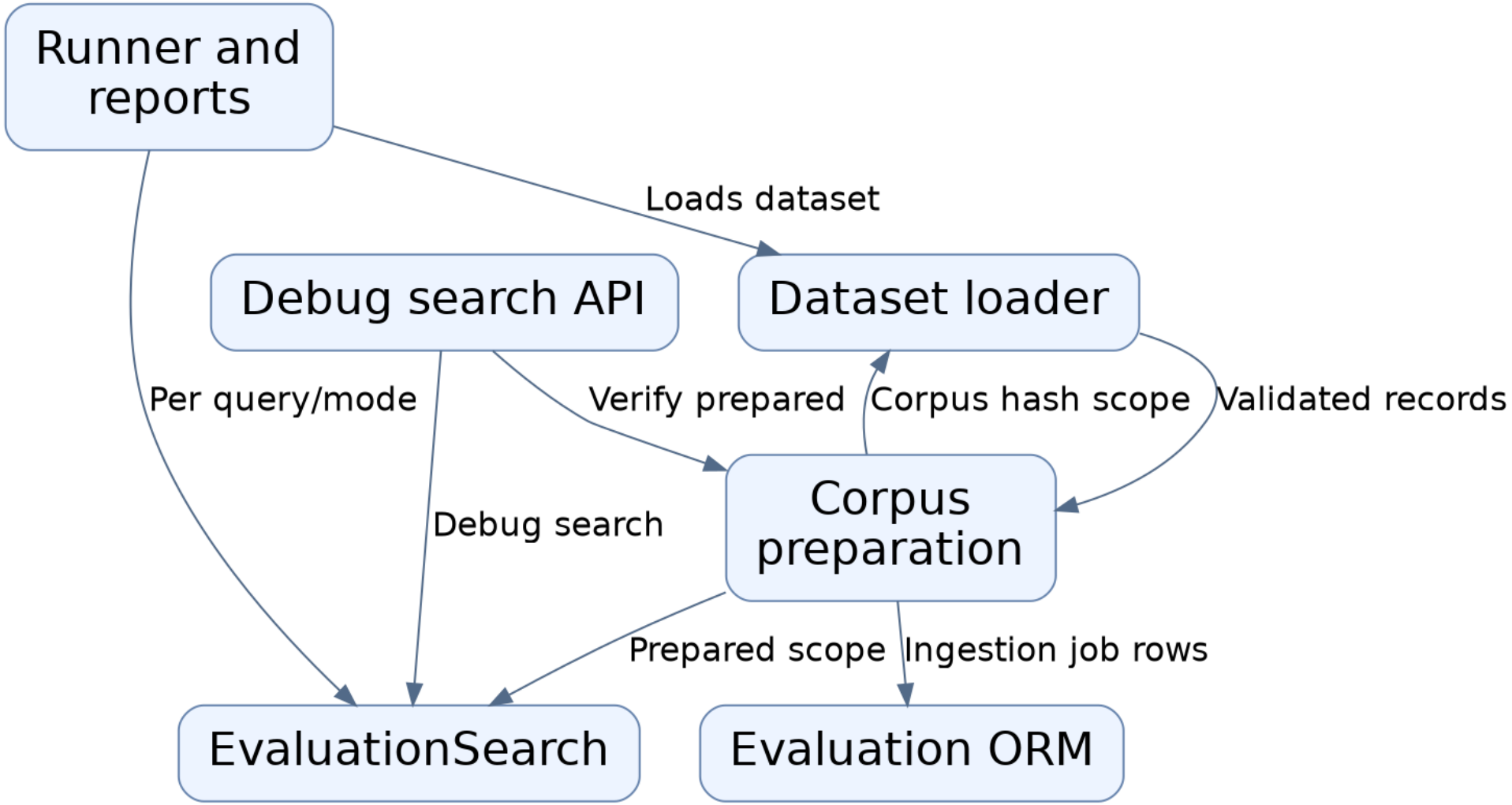
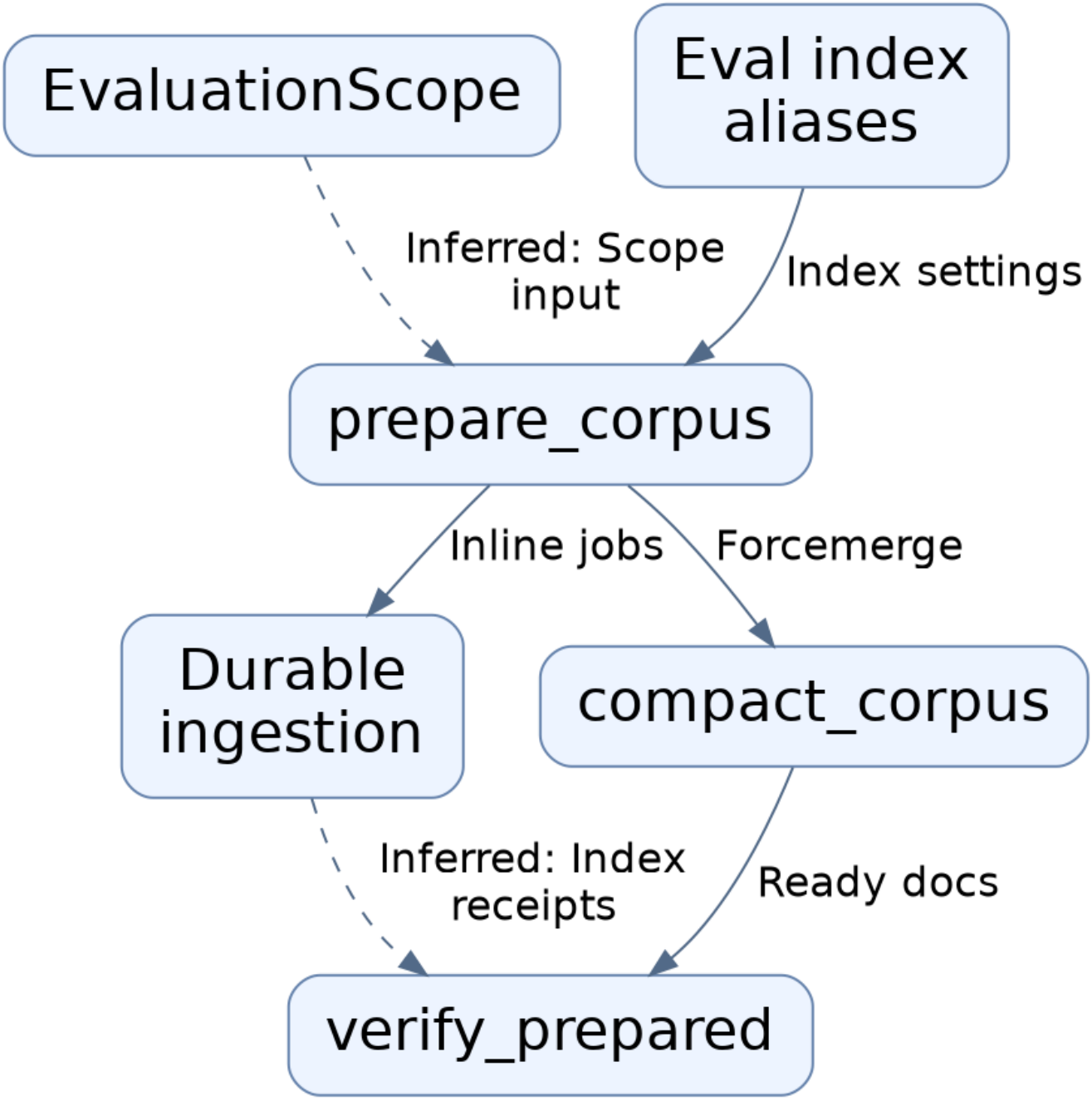
Architecture

Dataset loading, corpus preparation, search dispatch, scoring/reporting, and the HTTP surface are separate boundaries; the API and runner share one search boundary, while corpus state faults map to distinct errors.



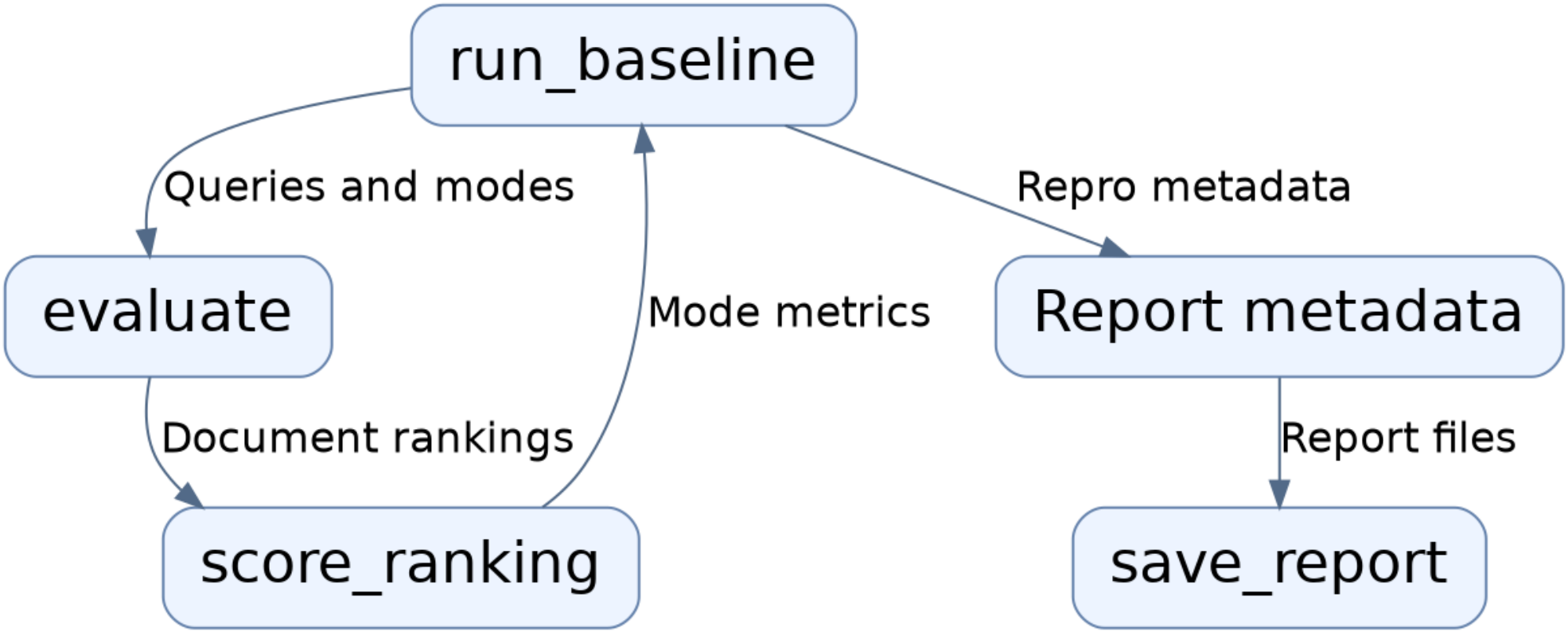
Corpus preparation flow

prepare_corpus derives a content-hash scope and isolated index aliases, runs durable ingestion inline, treats persisted documents as immutable, then force-merges and verifies against job receipts before search is allowed.



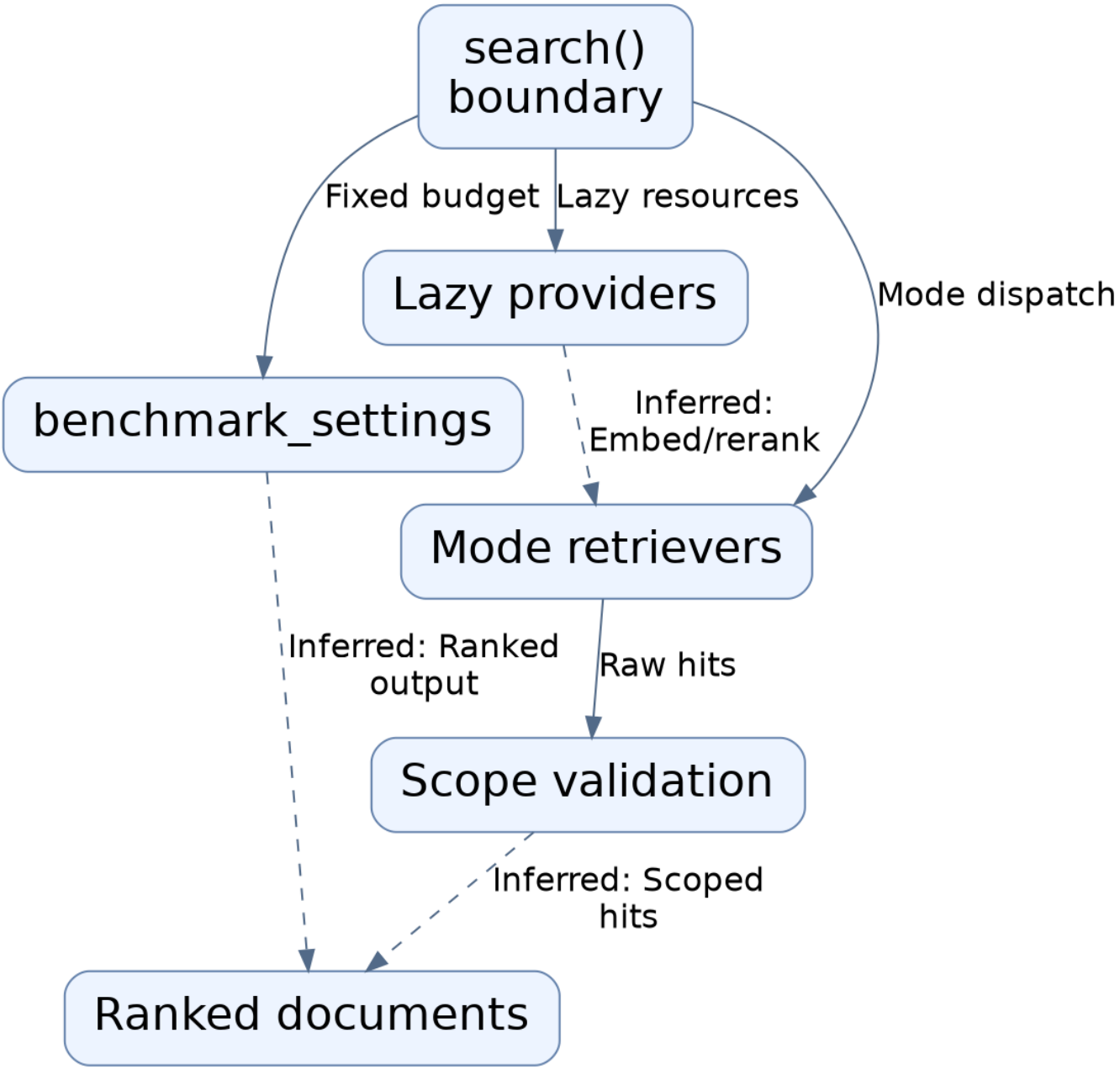
Evaluation run flow

run_baseline selects a dataset, verifies the alias, drives evaluate across queries and modes, scores rankings, attaches credential-free reproduction metadata, and writes JSON plus markdown reports, with bm25 as the delta reference.



Search mode dispatch

EvaluationSearch is the single dispatch boundary: it resolves dataset scope and aliases, applies the fixed benchmark budget, lazily builds only the needed retriever, and rejects results outside the prepared scope.



SciFact dataset build

The SciFact CLI verifies a pinned archive, converts BEIR files into the shared four-file contract, and builds a deterministic 30-query/500-document subset that load_dataset re-validates before registration.

